

TechBiz Academy

Week 2 · Task 5

Networking Tools & Subnetting Basics

ETHICAL HACKING INTERNSHIP

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Organization

TechBizz Security

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Introduction

Introduction

Computer networking is one of the most important areas of Cyber Security. Every ethical hacker must understand how computers communicate over a network before performing activities such as penetration testing, vulnerability assessment, or digital forensics.

A computer network allows devices to exchange information using Internet Protocol (IP). During communication, data travels through routers, switches, and servers before reaching its destination.

Network administrators and cybersecurity professionals use various diagnostic tools to identify network problems, verify connectivity, monitor active sessions, and analyze routing paths. These tools are also essential during the reconnaissance phase of ethical hacking because they provide information about target systems and network infrastructure.

In this laboratory experiment, networking utilities such as Ping, Traceroute, NSLookup, DIG, and Netstat were used. A subnetting exercise was also completed to understand how IPv4 networks are divided into smaller subnetworks.

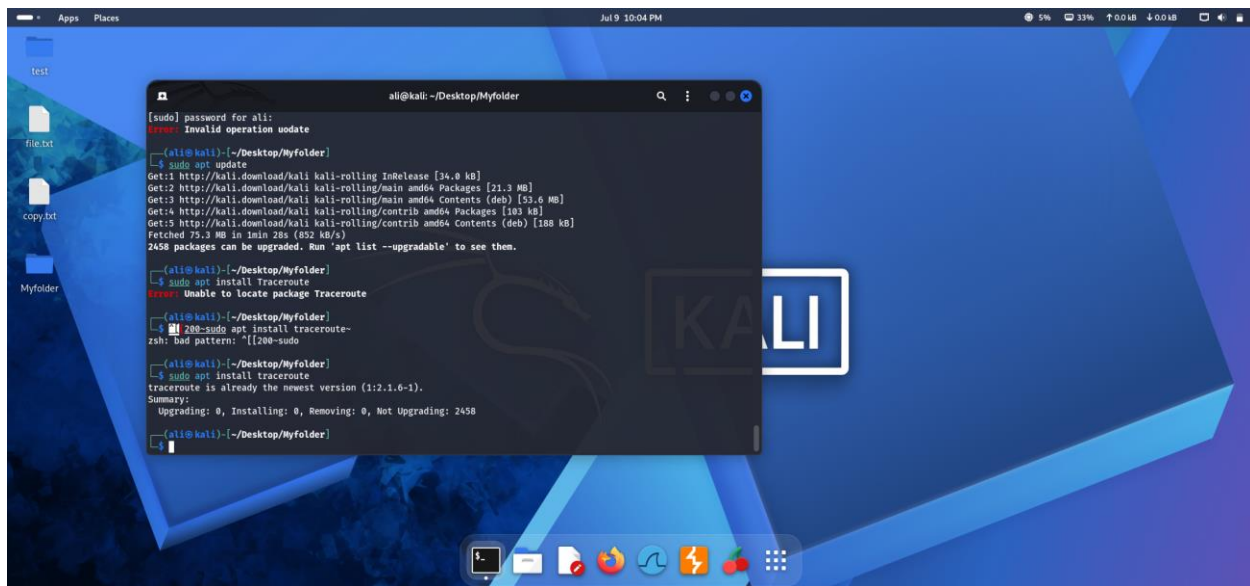


Figure 2: Basic Computer Network Diagram (PC → Router → ISP → Internet → Google Server)

Objectives

Objectives

After completing this experiment, students should be able to:

- Understand how ICMP works.
- Test internet connectivity.
- Measure network latency.
- Identify routing paths.
- Understand DNS resolution.
- View active network sessions.
- Learn subnetting concepts.
- Calculate usable host addresses.
- Understand CIDR notation.
- Apply networking concepts in cybersecurity.

Tools Required

- Windows 10/11 or Kali Linux

- Internet Connection
- Command Prompt
- Kali Linux Terminal
- DNS Server
- Google Chrome
- TryHackMe Platform



Figure 3: Windows Command Prompt or Kali Linux Terminal

Theory of Ping

What is Ping?

Ping is one of the oldest and most commonly used network troubleshooting tools.

Ping uses the **Internet Control Message Protocol (ICMP)** to determine whether another computer or server is reachable.

When the Ping command is executed, the computer sends an **ICMP Echo Request** packet to the destination. If the destination is online and reachable, it replies with an **ICMP Echo Reply** packet.

This process confirms that communication between the two devices is functioning correctly.

Ping is commonly used to:

- Check internet connectivity.
- Verify that a server is online.
- Measure network speed.
- Detect packet loss.
- Troubleshoot network issues.

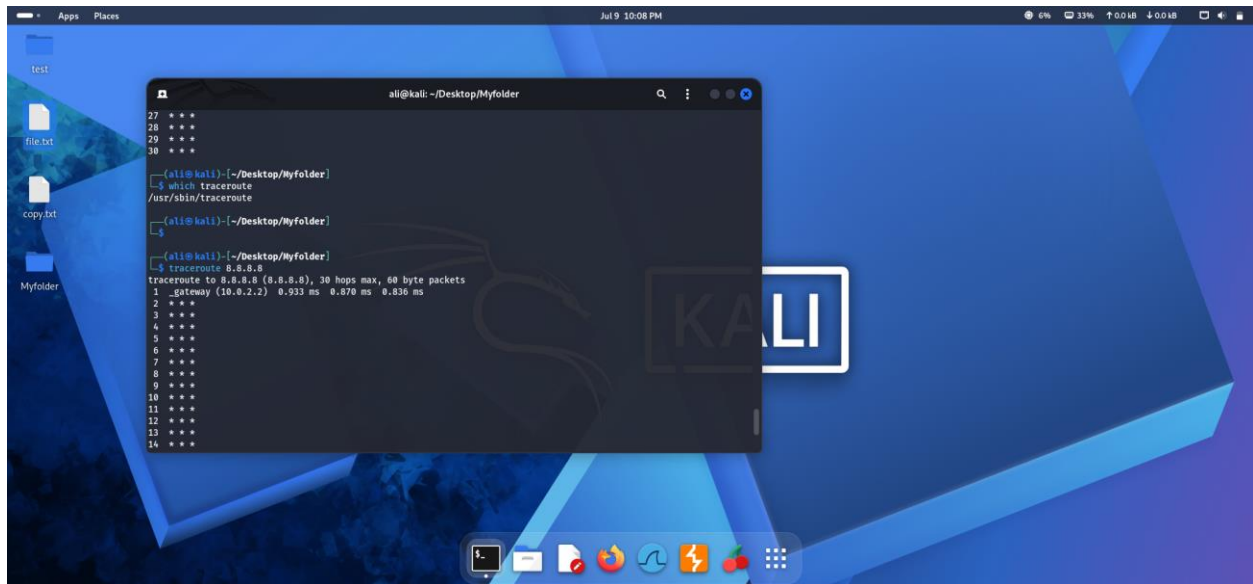


Figure 4: ICMP Echo Request and Echo Reply Diagram

Practical (Ping)

Command

```
ping google.com
```

Explanation

Explain each field separately:

Reply From

Indicates that the destination successfully responded.

Time

Represents Round Trip Time.

Example

22 ms

Lower values indicate faster communication.

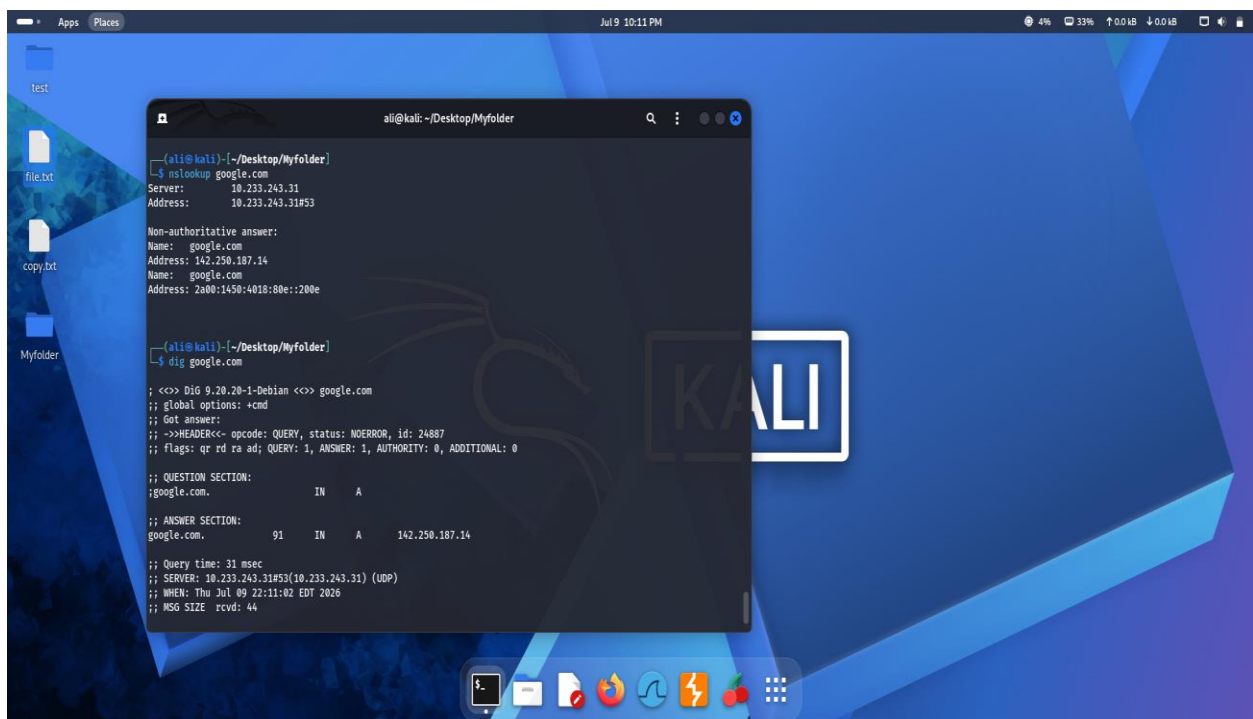
TTL

Time To Live prevents packets from looping forever.

Packet Loss

Shows how many packets were lost.

0% means an excellent connection.

A screenshot of a Kali Linux desktop environment. The desktop background is a blue and black geometric pattern with the word 'KALI' in a white box. A terminal window is open, showing the results of two network commands. The first command is 'nslookup google.com', which shows the server address as 10.233.243.31 and the IP address as 142.250.187.14. The second command is 'dig google.com', which shows detailed DNS information including the query time, server address, and the IP address 142.250.187.14. The terminal window title is 'ali@kali: ~/Desktop/Myfolder'. The desktop has a sidebar with icons for 'test', 'file.txt', 'copy.txt', and 'Myfolder'. The top status bar shows the date 'Jul 9 10:11 PM' and system resources like 4% CPU, 33% memory, and 0.0 KB network activity.

```
ali@kali: ~/Desktop/Myfolder
--(ali@kali)~/Desktop/Myfolder
--$ nslookup google.com
Server:      10.233.243.31
Address:     10.233.243.31#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.187.14
Name:   google.com
Address: 2a00:1450:4018:80e::200e

--(ali@kali)~/Desktop/Myfolder
--$ dig google.com

;; <<> Dig 9.20.20-1-Debian <<> google.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 24887
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;google.com.                IN      A

;; ANSWER SECTION:
google.com.                91      IN      A      142.250.187.14

;; Query time: 31 msec
;; SERVER: 10.233.243.31#53(10.233.243.31) (UDP)
;; WHEN: Thu Jul 09 22:11:02 EDT 2026
;; MSG SIZE rcvd: 44
```

Figure 5: Screenshot of Ping Output

Figure 6: Traceroute Working Diagram

Practical (Traceroute)

Command

```
tracert google.com
```

or

```
traceroute google.com
```

Explain:

Hop 1 → Local Router

Hop 2 → ISP

Hop 3 → ISP Backbone

Hop 4 → Google

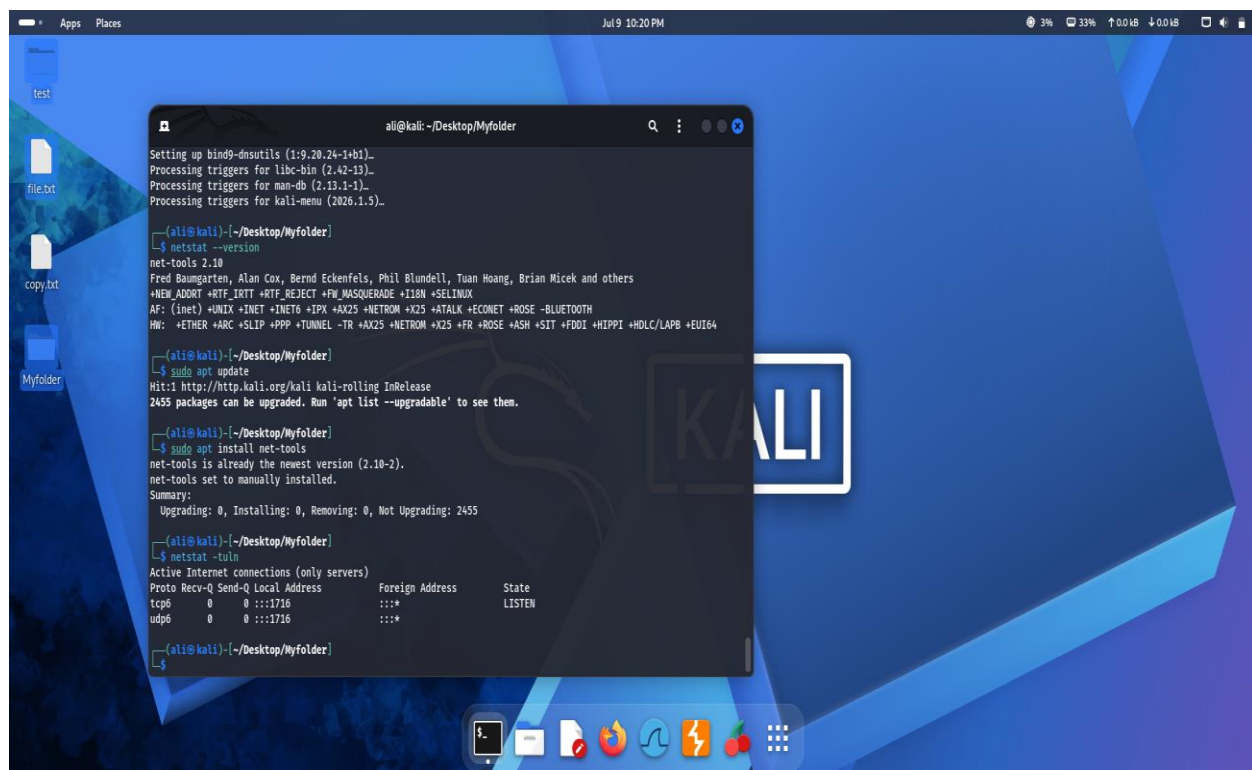


Figure 7: Screenshot of Traceroute Output

NSLookup

Explain:

DNS converts names into IP addresses.

Without DNS we would have to remember numerical IP addresses.

Example

google.com



142.250.xxx.xxx

Command

```
nslookup google.com
```

Explain

- Server
- Address
- DNS
- IP

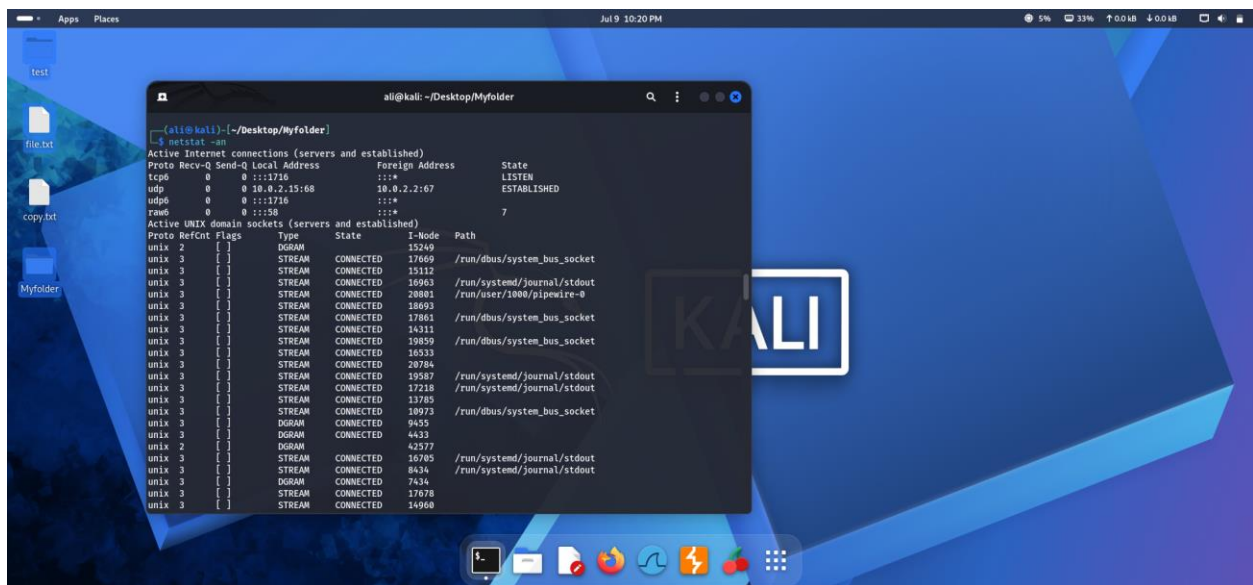


Figure 8: Screenshot of NSLookup Output

DIG

Explain

DIG means

Domain Information Groper

It provides

- TTL
- Query Time
- DNS Server
- Record Type
- Answer Section

Command

`dig google.com`

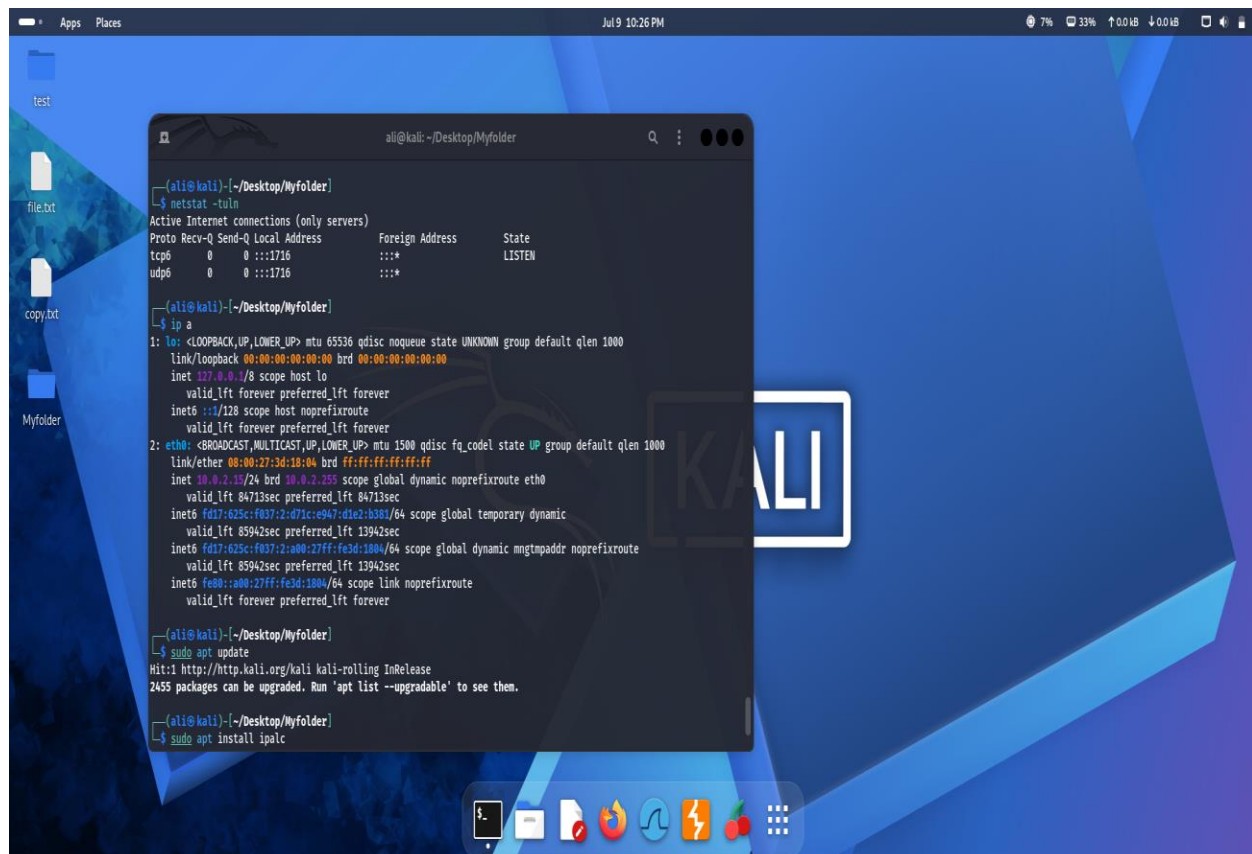


Figure 9: Screenshot of DIG Output

Netstat

Explain

Netstat displays

- Active connections
- Listening ports
- Routing table
- Protocol statistics

Command

netstat

Explain

TCP

UDP

Foreign Address

State

Listening

Established



Figure 10: Screenshot of Netstat Output

Subnetting Theory

Explain

IPv4 Address

Network ID

Host ID

CIDR

Subnet Mask

Formula

$\text{Hosts} = 2^n - 2$

Where

n = Host Bits

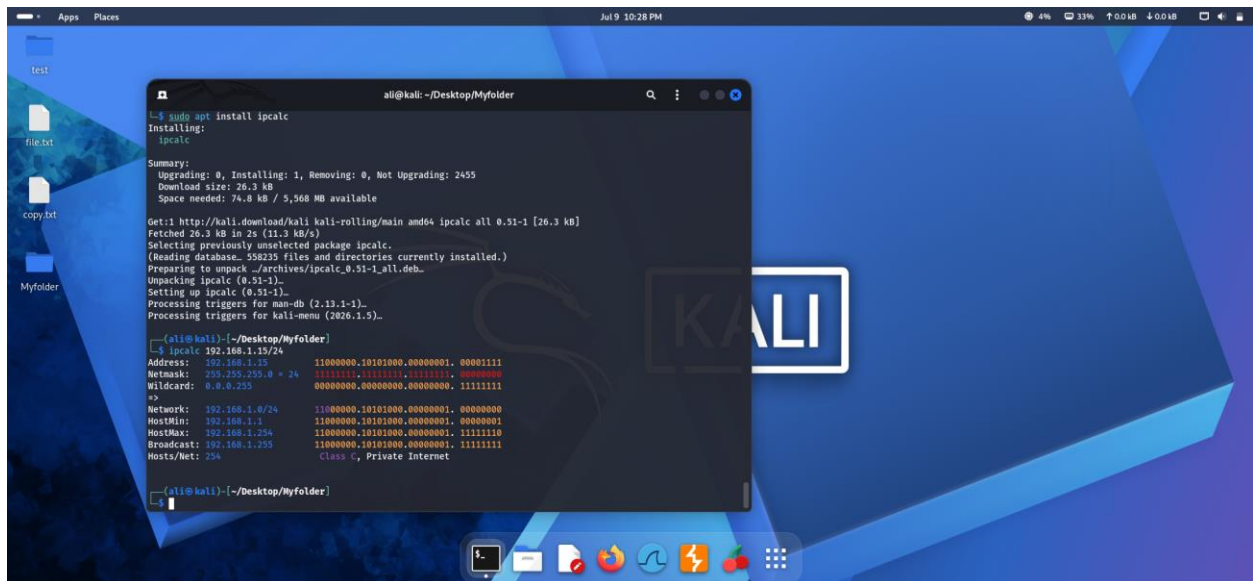


Figure 11: IPv4 Address Structure (32 bits split into Network ID and Host ID)

Practical Subnetting

Example

192.168.10.0/24

Network Address

192.168.10.0

First Host

192.168.10.1

Last Host

192.168.10.254

Broadcast

192.168.10.255

Usable Hosts

25

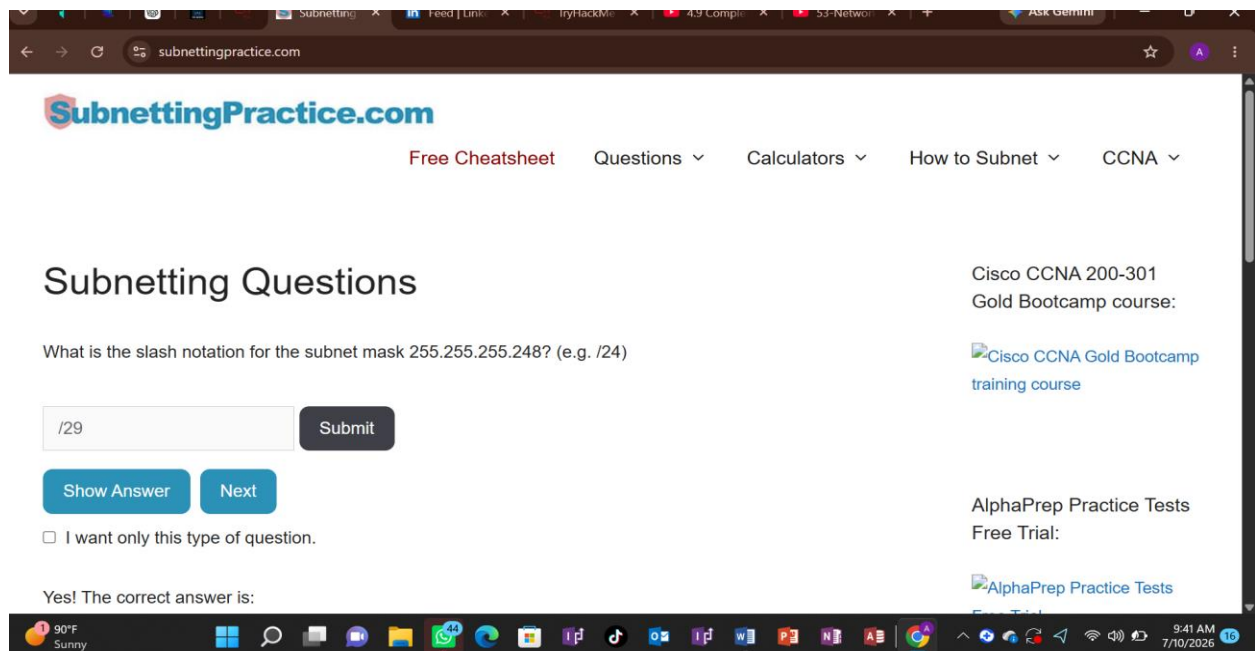


Figure 12: Subnetting Diagram

Observations

- Internet connection was stable.
- No packet loss was observed during the Ping test.
- Traceroute successfully identified all network hops.
- DNS resolution correctly translated the domain name into an IP address.
- Netstat displayed active TCP connections and listening ports.
- The subnetting calculation confirmed that a /24 network supports **254 usable host addresses**.

Applications

These tools are widely used in:

- Ethical Hacking
- Penetration Testing
- Vulnerability Assessment
- Network Administration
- Digital Forensics
- Security Operations Center (SOC)
- Incident Response
- Cloud Networking
- Cybersecurity Monitoring

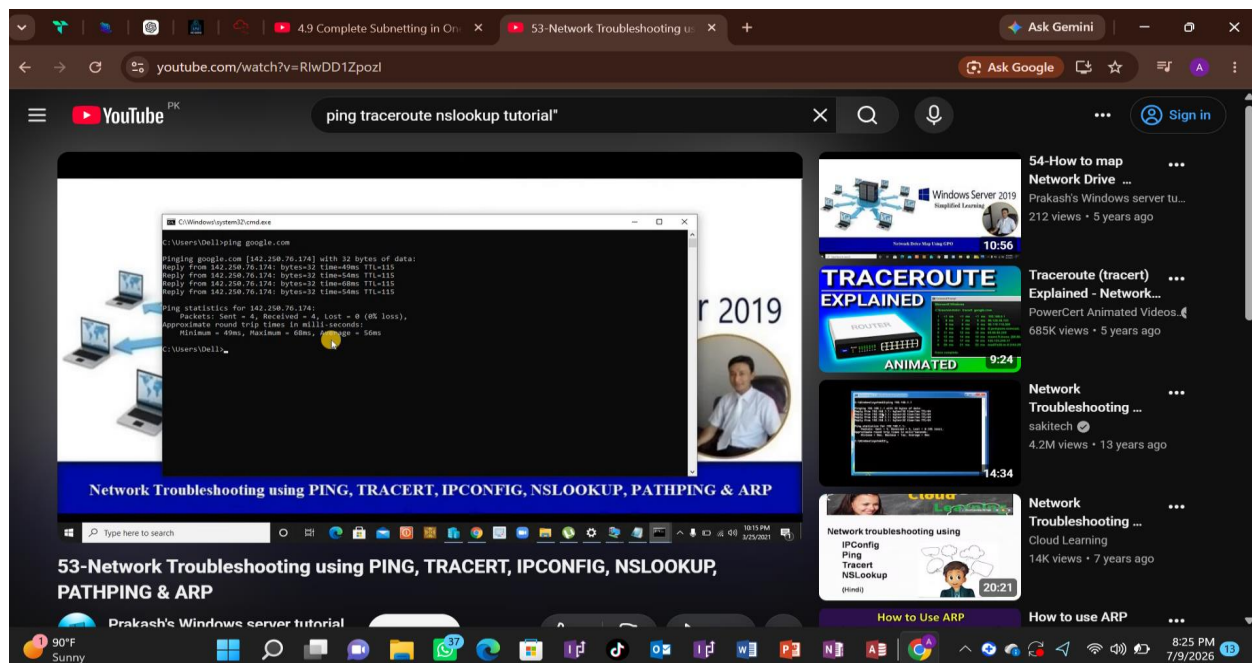


Figure 13: Cybersecurity Operations Center (SOC) Illustration

Advantages

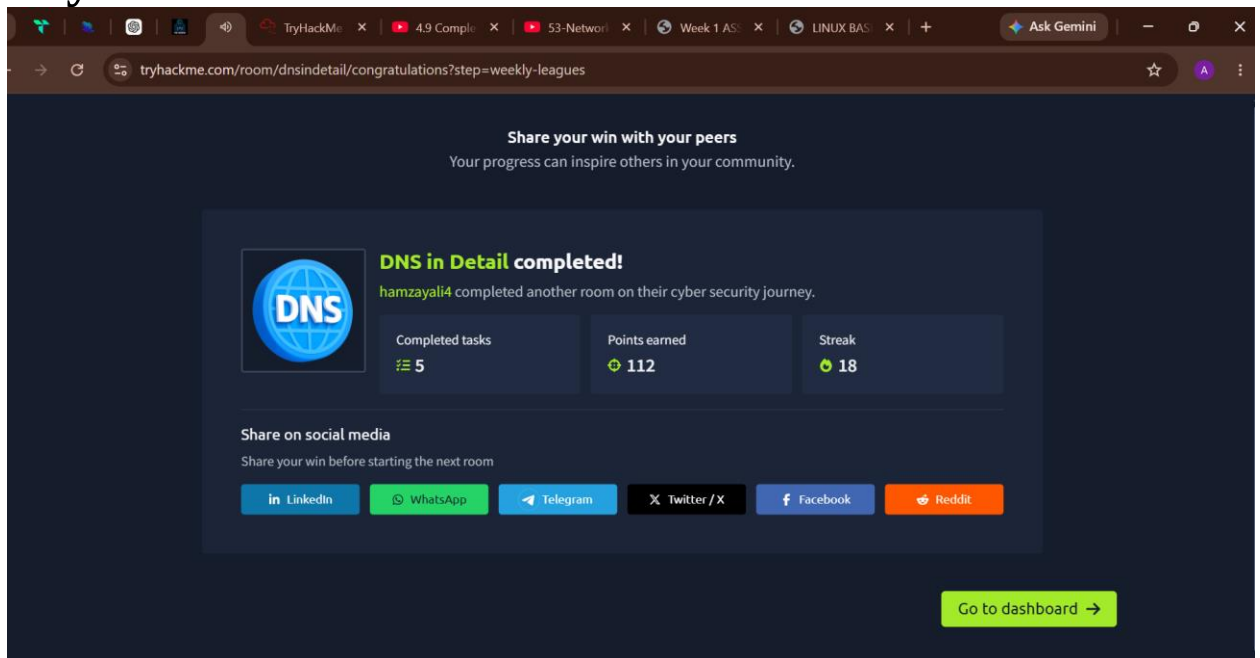
- Easy to use
- Fast troubleshooting
- Helps detect connectivity issues
- Improves network understanding
- Supports ethical hacking tasks
- Assists in DNS analysis
- Helps identify suspicious network activity

Conclusion

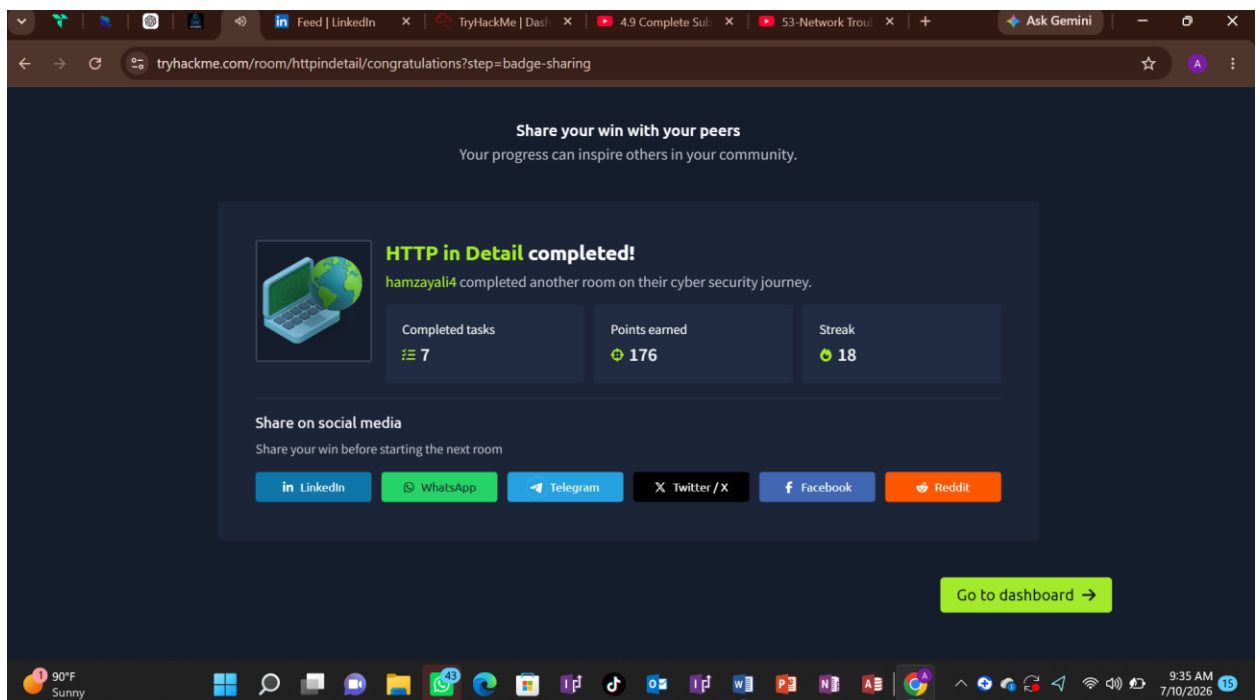
This experiment provided practical experience with essential networking utilities used in cybersecurity and ethical hacking. By using Ping, Traceroute, NSLookup, DIG, and Netstat, network connectivity, routing paths, DNS resolution, and active sessions were analyzed successfully. The subnetting exercise also strengthened understanding of IPv4 addressing and CIDR notation. Overall, the experiment demonstrated the importance of these tools in diagnosing network issues, conducting reconnaissance, and supporting secure network administration.

References

• TryHackMe – DNS in Detail



TryHackMe – HTTP in Detail



SubnettingPractice.com

The screenshot shows the SubnettingPractice.com website. The header includes the site name and navigation links: Free Cheatsheet, Questions, Calculators, How to Subnet, and CCNA. The main content area is titled "Subnetting Questions" and contains a question: "What is the slash notation for the subnet mask 255.255.255.248? (e.g. /24)". Below the question is an input field with the value "/29" and a "Submit" button. There are also buttons for "Show Answer" and "Next". A checkbox option reads "I want only this type of question." Below the question, it says "Yes! The correct answer is:". On the right side, there are links for "Cisco CCNA 200-301 Gold Bootcamp course" and "AlphaPrep Practice Tests Free Trial". The browser's taskbar at the bottom shows the date as 7/10/2026 and the time as 9:41 AM.

• Video: "subnetting explained for beginners"

The screenshot shows a YouTube video player. The video title is "4.9 Complete Subnetting in One video" by "KnowledgeGATE by Sanchit Sir". The video content shows a man speaking in front of a blue background with a diagram of houses and a network diagram. The video has 9.6K likes and a "Share" button. On the right side, there are several video recommendations, including "UNICAST BROADCAST MULTICAST", "IP ADDRESSING", "4.8 Types of Casting in Computer", "4.6 What is Logical or IP Addressing", "Live Subnetting Class in Hindi", and "Subnetting by Hand (like a Boss)". The browser's taskbar at the bottom shows the date as 7/9/2026 and the time as 7:51 PM.

.Video: "ping traceroute ns lookup tutorial"

The screenshot shows a YouTube video player with the title "53-Network Troubleshooting using PING, TRACERT, IPCONFIG, NSLOOKUP, PATHPING & ARP". The video content displays a Windows command prompt window with the following output:

```
C:\Users\ DELL>ping google.com

Pinging google.com [142.250.76.174] with 32 bytes of data:
Reply from 142.250.76.174: bytes=32 time=49ms TTL=115
Reply from 142.250.76.174: bytes=32 time=54ms TTL=115
Reply from 142.250.76.174: bytes=32 time=48ms TTL=115
Reply from 142.250.76.174: bytes=32 time=54ms TTL=115

Ping statistics for 142.250.76.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 49ms, Maximum = 68ms, Average = 56ms
```

The video player interface includes a search bar with the text "ping traceroute nslookup tutorial", a "Sign in" button, and a sidebar with recommended videos. The taskbar at the bottom shows the Windows Start button, search bar, and various application icons. The system tray indicates a temperature of 90°F and a sunny weather condition.

THANK
YOU! 😊